

21 Compress and decompress files using tar.

```
chunduru@chunduru-virtual-machine:~$ mkdir test
chunduru@chunduru-virtual-machine:~$ cd test
chunduru@chunduru-virtual-machine:~/test$ touch file1 file2
chunduru@chunduru-virtual-machine:~/test$ ls
file1 file2
chunduru@chunduru-virtual-machine:~/test$ cd ..
chunduru@chunduru-virtual-machine:~$ tar -cvf test.tar test
test/
test/file2
test/file1
chunduru@chunduru-virtual-machine:~$ gzip test.tar
chunduru@chunduru-virtual-machine:~$ ls
Desktop Documents Downloads file.text karthik linl.text Music Pictures Public snap techie.tar Templates test test-tar test.tar.gz Videos
chunduru@chunduru-virtual-machine:~$ gunzip test.tar.gz
chunduru@chunduru-virtual-machine:~$ ls
Desktop Documents Downloads file.text karthik linl.text Music Pictures Public snap techie.tar Templates test test-tar test.tar Videos
chunduru@chunduru-virtual-machine:~$ mkdir extract
chunduru@chunduru-virtual-machine:~$ tar -xvf test.tar -C extract
tar: You may not specify more than one '-Acdrux', '--delete' or '--test-label' option
Try 'tar --help' or 'tar --usage' for more information.
chunduru@chunduru-virtual-machine:~$ tar -xvf test.tar -C extract
test/
test/file2
test/file1
chunduru@chunduru-virtual-machine:~$ tar -cvf test.tar.gz
tar: Cowardly refusing to create an empty archive
Try 'tar --help' or 'tar --usage' for more information.
chunduru@chunduru-virtual-machine:~$ tar -czvf test.tar.gz test
test/
test/file2
test/file1
chunduru@chunduru-virtual-machine:~$ rm -rf test
chunduru@chunduru-virtual-machine:~$ ls
Desktop Documents Downloads extract file.text karthik linl.text Music Pictures Public snap techie.tar Templates test test-tar test.tar test.tar.gz Videos
chunduru@chunduru-virtual-machine:~$ tar -xzvf test.tar.gz
test/
test/file2
test/file1
chunduru@chunduru-virtual-machine:~$ ls
Desktop Documents Downloads extract file.text karthik linl.text Music Pictures Public snap techie.tar Templates test test-tar test.tar test.tar.gz Videos
chunduru@chunduru-virtual-machine:~$
```

1. Purpose

The primary goal of this terminal session is to demonstrate how to manage compressed archives in Linux using tar and gzip. It covers creating directories, archiving files, compressing them into .tar.gz format, handling extraction errors, and restoring files directly to specific folders.

2. Command Execution

The session executes a series of core Linux commands to achieve this:

- **mkdir & touch:** Created a test directory containing sample files (file1, file2).
- **tar -cvf:** Combined the files into an uncompressed test.tar archive.
- **gzip & gunzip:** Compressed test.tar into test.tar.gz and decompressed it back manually.
- **tar -xvf ... -C extract:** Extracted files into a specific target folder (extract).
- **tar -czvf & tar -xzvf:** Created and extracted a gzip-compressed archive in single-step commands using the -z flag.

3. Verification & Error Handling

Each action is verified immediately using the ls command to inspect file creation, file renaming, and directory contents. The log also illustrates common user errors, such as misplacing flags (tar -xvf test.tar -c extract) and trying to archive a non-existent folder after deleting it (rm -rf test), which produces the error tar: Cowardly refusing to create an empty archive.

4. Conclusion

The session proves that while archiving (tar) and compressing (gzip) can be done in separate manual steps, the standard best practice in Linux is to use single-step flags: tar -czvf for compression and tar -xzvf (or tar -xvf) for extraction.

22 . Monitor system resources with top or htop.

```
chundururuchundurur-virtual-machine:~$ top

top - 17:35:15 up 6:46, 1 user, load average: 0.04, 0.05, 0.07
Tasks: 319 total, 1 running, 318 sleeping, 0 stopped, 0 zombie
%Cpu(s): 7.0 us, 2.7 sy, 0.0 ni, 89.6 id, 0.0 wa, 0.0 hi, 0.7 si, 0.0 st
MiB Mem : 1920.3 total, 180.3 free, 1070.3 used, 669.7 buff/cache
MiB Swap: 2140.0 total, 1410.1 free, 729.9 used. 634.4 avail Mem

  PID USER      PR  NI    VIRT    RES    SHR  S  %CPU  %MEM    TIME+  COMMAND
 1640 chundurur 20   0 4476272 172404 68160  S   8.6   8.8   5:20.14 gnome-shell
 5755 chundurur 20   0 3516804 315844 114868  S   5.6  16.1   2:49.45 firefox
 2308 chundurur 20   0 562956   34820 25284  S   3.0   1.8   1:26.83 gnome-terminal-
 6810 chundurur 20   0 2404136 61292 52720  S   0.7   3.1   0:06.23 Isolated Web Co
11635 chundurur 20   0 13340 4348 3368  R   0.7   0.2   0:01.06 top
 561 systemd+ 20   0 14836 4460 4320  S   0.3   0.2   1:03.65 systemd-oomd
 594 root      20   0 317384 5724 5188  S   0.3   0.3   1:55.01 vmttoolsd
1838 chundurur 20   0 144344 19044 15856  S   0.3   1.0   1:48.04 vmttoolsd
   1 root      20   0 168100 10712 6340  S   0.0   0.5   0:11.82 systemd
   2 root      20   0 0 0 0  S   0.0   0.0   0:00.17 kthreadd
   3 root      20   0 0 0 0  S   0.0   0.0   0:00.00 pool_workqueue_release
   4 root      0 -20 0 0 0  I   0.0   0.0   0:00.00 kworker/R-rcu_g
   5 root      0 -20 0 0 0  I   0.0   0.0   0:00.00 kworker/R-rcu_p
   6 root      0 -20 0 0 0  I   0.0   0.0   0:00.00 kworker/R-slub_
   7 root      0 -20 0 0 0  I   0.0   0.0   0:00.00 kworker/R-netns
  11 root      20   0 0 0 0  I   0.0   0.0   0:00.00 kworker/u256:0-ipv6_addrconf
  12 root      0 -20 0 0 0  I   0.0   0.0   0:00.00 kworker/R-mm_pe
  13 root      20   0 0 0 0  I   0.0   0.0   0:00.00 rcu_tasks_kthread
  14 root      20   0 0 0 0  I   0.0   0.0   0:00.00 rcu_tasks_rude_kthread
  15 root      20   0 0 0 0  I   0.0   0.0   0:00.00 rcu_tasks_trace_kthread
  16 root      20   0 0 0 0  S   0.0   0.0   0:00.89 ksoftirqd/0
  17 root      20   0 0 0 0  I   0.0   0.0   0:04.27 rcu_preempt
  18 root      rt   0 0 0 0  S   0.0   0.0   0:00.26 migration/0
  19 root     -51  0 0 0 0  S   0.0   0.0   0:00.00 idle_inject/0
  20 root      20   0 0 0 0  S   0.0   0.0   0:00.00 cpuhp/0
  21 root      20   0 0 0 0  S   0.0   0.0   0:00.00 cpuhp/1
  22 root     -51  0 0 0 0  S   0.0   0.0   0:00.00 idle_inject/1
  23 root      rt   0 0 0 0  S   0.0   0.0   0:00.94 migration/1
  24 root      20   0 0 0 0  S   0.0   0.0   0:00.92 ksoftirqd/1
  29 root      20   0 0 0 0  S   0.0   0.0   0:00.00 kdevtmpfs

0% ||| 1.3%| Tasks: 136, 469 thr; 1 running
1% ||| 3.9%| Load average: 0.76 0.31 0.16
Mem[|||||] 1.89G/2.88G| Uptime: 06:49:52
Swp[|||||] 730M/2.89G|

  PID USER      PRI  NI    VIRT    RES    SHR  S  %CPU  %MEM    TIME+  Command
 1640 chundurur 20   0 4371M 171M 70060  S   0.7  9.0   5:25.12 /usr/bin/gnome-shell
 2308 chundurur 20   0 549M 35484 25764  S   0.0  1.8   1:29.74 /usr/libexec/gnome-terminal-server
12100 chundurur 20   0 12020 5712 5612  R   2.6  0.3   0:00.81 htop
 904 root      20   0 73252 7388 8852  S   0.0  0.4   0:00.74 /usr/sbin/cupsd -l
 594 root      20   0 309M 5660 5176  S   0.7  0.3   1:55.79 /usr/bin/vmttoolsd
1667 chundurur 20   0 4371M 171M 70060  S   0.0  9.0   0:05.62 /usr/bin/gnome-shell
1838 chundurur 20   0 140M 22520 10336  S   0.7  1.1   1:48.90 /usr/bin/vmttoolsd -n vmusr --blockFd 3 --uinputFd 4
   1 root      20   0 164M 10576 6204  S   0.0  0.5   0:11.98 /sbin/init auto noprompt splash
 334 root     19 -1 64624 18048 17356  S   0.0  0.9   0:06.23 /lib/systemd/systemd-journald
 365 root      20   0 148M 1360 1220  S   0.0  0.1   0:00.02 vmware-vmblock-fuse /run/vmblock-fuse -o rw,subtype=vmware-vmblock,default_permissions,allow_other,dev,suid
 366 root      20   0 148M 1360 1220  S   0.0  0.1   0:00.02 vmware-vmblock-fuse /run/vmblock-fuse -o rw,subtype=vmware-vmblock,default_permissions,allow_other,dev,suid
 367 root      20   0 148M 1360 1220  S   0.0  0.1   0:00.00 vmware-vmblock-fuse /run/vmblock-fuse -o rw,subtype=vmware-vmblock,default_permissions,allow_other,dev,suid
 384 root      20   0 27192 3664 3388  S   0.0  0.2   0:00.96 /lib/systemd/systemd-udev
 561 systemd-o 20   0 14836 4388 4248  S   0.0  0.2   1:04.08 /lib/systemd/systemd-oomd
 563 systemd-r 20   0 26736 6652 5268  S   0.0  0.3   0:03.33 /lib/systemd/systemd-resolved
 565 systemd-t 20   0 89388 4880 4752  S   0.0  0.2   0:00.46 /lib/systemd/systemd-timesyncd
 581 systemd-t 20   0 89388 4880 4752  S   0.0  0.2   0:00.00 /lib/systemd/systemd-timesyncd
 592 root      20   0 53756 4272 4268  S   0.0  0.2   0:00.09 /usr/bin/VGAuthService
 684 root      20   0 309M 5660 5176  S   0.0  0.3   0:01.95 /usr/bin/vmttoolsd
 685 root      20   0 309M 5660 5176  S   0.0  0.3   0:00.00 /usr/bin/vmttoolsd
 689 root      20   0 234M 7084 6496  S   0.0  0.4   0:01.45 /usr/libexec/accounts-daemon
 691 root      20   0 2816 2036 1872  S   0.0  0.1   0:01.42 /usr/sbin/acpid
 695 root      20   0 309M 5660 5176  S   0.0  0.3   0:00.00 /usr/bin/vmttoolsd
 698 avahi      20   0 7636 3220 3032  S   0.0  0.2   0:01.13 avahi-daemon: running [chundurur-virtual-machine.local]
 699 root      20   0 9496 2788 2680  S   0.0  0.1   0:00.20 /usr/sbin/cron -f -P
 702 messagebu 20   0 10996 5180 3292  S   0.0  0.3   0:03.38 @dbus-daemon --system --address=systemd: --systemd-activation --syslog-only
 707 root      20   0 399M 12828 10736  S   0.0  0.7   0:04.58 /usr/sbin/NetworkManager --no-daemon
 723 root      20   0 82840 3688 3520  S   0.0  0.2   0:02.96 /usr/sbin/NetworkManager --foreground
 727 root      20   0 41176 6336 6332  S   0.0  0.3   0:00.32 /usr/bin/python3 /usr/bin/networkd-dispatcher --run-startup-triggers
 731 root      20   0 237M 8336 7076  S   0.0  0.4   0:03.78 /usr/libexec/polkitd --no-debug
 733 root      20   0 234M 7084 6496  S   0.0  0.4   0:01.00 /usr/libexec/accounts-daemon
 736 root      20   0 234M 8336 6332  S   0.0  0.3   0:00.06 /usr/libexec/power-profiles-daemon
 740 syslog    20   0 217M 3820 3380  S   0.0  0.2   0:00.49 /usr/sbin/rsyslogd -n -lNONE
 742 root      20   0 237M 8336 7076  S   0.0  0.4   0:00.00 /usr/libexec/polkitd --no-debug

~?Help F2Setup F3Search F4Filter F5Tree F6SortBy F7Nice F8NICE F9Kill F10Quit
```

1. Purpose

top and htop are real-time monitoring tools used in Linux to track CPU, memory, load averages, and running processes to identify system bottlenecks.

2. Command Execution & Verification

top provides a standard text-based summary of running tasks and resource consumption, while htop offers an interactive, color-coded visual interface to search, filter, and manage processes easily.

3. Conclusion

Both utilities verify system performance and process activity, but htop is preferred for interactive troubleshooting while top is the essential default available on all Linux systems.

23 .Create and manage user groups.

```
chunduru@chunduru-virtual-machine:~$ sudo groupadd developers
chunduru@chunduru-virtual-machine:~$ getent group developers
developers:x:1002:
chunduru@chunduru-virtual-machine:~$ sudo usermod -aG developers karthik
usermod: user 'karthik' does not exist
chunduru@chunduru-virtual-machine:~$ sudo usermod -aG developers chunduru
chunduru@chunduru-virtual-machine:~$ whoami
chunduru
chunduru@chunduru-virtual-machine:~$ groups
chunduru adm cdrom sudo dip plugdev lpadmin lxd sambashare
chunduru@chunduru-virtual-machine:~$ groups chunduru
chunduru : chunduru adm cdrom sudo dip plugdev lpadmin lxd sambashare de
chunduru@chunduru-virtual-machine:~$ getent group developers
command 'getent' not found, did you mean:
  command 'getent' from deb libc-bin (2.35-0ubuntu3.13)
Try: sudo apt install <deb name>
chunduru@chunduru-virtual-machine:~$ getent group developers
developers:x:1002:chunduru
chunduru@chunduru-virtual-machine:~$
```

Purpose

Managing user groups in Linux involves creating new group accounts and adding existing users to them so permissions can be controlled collectively.

Command Execution and Verification

The command `sudo groupadd developers` creates a new group named `developers`, which is verified using `getent group developers`. Attempting `sudo usermod -aG developers karthik` fails because the user `karthik` does not exist, but running `sudo usermod -aG developers chunduru` succeeds and adds user `chunduru` to the group. Running `groups chunduru` and `getent group developers` verifies that `chunduru` is now a member of the `developers` group.

Conclusion

Linux group management relies on `groupadd` to define new access roles and `usermod` with the `-aG` flags to attach users to these groups without overwriting their existing group memberships.

24 .Set up SSH password less authentication.

```
chundur@chundur-virtual-machine:~$ ssh-keygen
Generating public/private rsa key pair.
Enter file in which to save the key (/home/chundur/.ssh/id_rsa):
/home/chundur/.ssh/id_rsa already exists.
Overwrite (y/n)? Y
Enter passphrase (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in /home/chundur/.ssh/id_rsa
Your public key has been saved in /home/chundur/.ssh/id_rsa.pub
The key fingerprint is:
SHA256:skoR08n6VbHzmQAvnoSLThAcODcu2F6Xbpj2UFHY9fQ chundur@chundur-virtual-machine
The key's randomart image is:
+---[RSA 3072]-----+
| oo. .+o.. . |
| o +. o ooo.o o . |
| ..oo=.o.* . E |
| o o.o==+.+ + o |
| o .+=+o.S + |
| .o o++ |
| o.oo |
| . . . |
| . |
+---[SHA256]-----+
chundur@chundur-virtual-machine:~$ ssh-copy-id chundur@192.168.1.100
/usr/bin/ssh-copy-id: INFO: Source of key(s) to be installed: "/home/chundur/.ssh/id_rsa.pub"
/usr/bin/ssh-copy-id: INFO: attempting to log in with the new key(s), to filter out any that are already installed
/usr/bin/ssh-copy-id: ERROR: ssh: connect to host 192.168.1.100 port 22: Connection refused

chundur@chundur-virtual-machine:~$ ssh chundur@REMOTE_IP
ssh: Could not resolve hostname remote_ip: Temporary failure in name resolution
chundur@chundur-virtual-machine:~$
```

Purpose

Monitoring log files with commands like tail or grep allows real-time tracking of system events and quick filtering of error messages or specific log entries.

Command Execution and Verification

Using tail -f /var/log/syslog actively streams new log lines as they are generated, while grep -i "error" /var/log/syslog searches through the log file to display only lines containing error conditions. Combining both using tail -f /var/log/syslog | grep -i "error" verifies live log activity while filtering out unnecessary noise.

Conclusion

Effective log monitoring relies on tail to follow dynamic updates and grep to extract critical diagnostic information instantly.

25. Monitor log files using tail or grep.

```

chundurur@chundurur-virtual-machine:~$ sudo tail -f /var/log/syslog
[sudo] password for chundurur:
Aug 8 17:35:56 chundurur-virtual-machine systemd[1]: esm-cache.service: Deactivated successfully.
Aug 8 17:35:56 chundurur-virtual-machine systemd[1]: Finished Update the local ESM caches.
Aug 8 17:35:57 chundurur-virtual-machine systemd[1]: esm-cache.service: Consumed 1.416s CPU time.
Aug 8 17:37:42 chundurur-virtual-machine dbus-daemon[1525]: [session uid=1000 pid=1525] Activating via systemd: service name='org.freedesktop.Tracker3.Miner.Extract' unit='tracker-extract-3.service' requested by '1.81' (uid=1000 pid=2067 comm="/usr/libexec/tracker-miner-fs-3" label="unconfined")
Aug 8 17:37:42 chundurur-virtual-machine systemd[1491]: Starting Tracker metadata extractor...
Aug 8 17:37:46 chundurur-virtual-machine dbus-daemon[1525]: [session uid=1000 pid=1525] Successfully activated service 'org.freedesktop.Tracker3.Miner.Extract'
Aug 8 17:37:56 chundurur-virtual-machine systemd[1491]: tracker-extract-3.service: Consumed 2.255s CPU time.
Aug 8 17:39:20 chundurur-virtual-machine NetworkManager[707]: <Info> [1786190960.5109] dhcp4 (ens33): state changed new lease, address=192.168.85.128
Aug 8 17:54:20 chundurur-virtual-machine NetworkManager[707]: <Info> [1786191860.5216] dhcp4 (ens33): state changed new lease, address=192.168.85.128
chundurur@chundurur-virtual-machine:~$ sudo tail -f /var/log/syslog
Aug 8 17:35:56 chundurur-virtual-machine systemd[1]: esm-cache.service: Deactivated successfully.
Aug 8 17:35:56 chundurur-virtual-machine systemd[1]: Finished Update the local ESM caches.
Aug 8 17:35:57 chundurur-virtual-machine systemd[1]: esm-cache.service: Consumed 1.416s CPU time.
Aug 8 17:37:42 chundurur-virtual-machine dbus-daemon[1525]: [session uid=1000 pid=1525] Activating via systemd: service name='org.freedesktop.Tracker3.Miner.Extract' unit='tracker-extract-3.service' requested by '1.81' (uid=1000 pid=2067 comm="/usr/libexec/tracker-miner-fs-3" label="unconfined")
Aug 8 17:37:42 chundurur-virtual-machine systemd[1491]: Starting Tracker metadata extractor...
Aug 8 17:37:46 chundurur-virtual-machine dbus-daemon[1525]: [session uid=1000 pid=1525] Successfully activated service 'org.freedesktop.Tracker3.Miner.Extract'
Aug 8 17:37:56 chundurur-virtual-machine systemd[1491]: tracker-extract-3.service: Consumed 2.255s CPU time.
Aug 8 17:39:20 chundurur-virtual-machine NetworkManager[707]: <Info> [1786190960.5109] dhcp4 (ens33): state changed new lease, address=192.168.85.128
Aug 8 17:54:20 chundurur-virtual-machine NetworkManager[707]: <Info> [1786191860.5216] dhcp4 (ens33): state changed new lease, address=192.168.85.128
^C
chundurur@chundurur-virtual-machine:~$ sudo grep -i "error" /var/log/syslog
Aug 4 11:15:45 chundurur-virtual-machine systemd[1]: Condition check resulted in Process error reports when automatic reporting is enabled (file watch) being skipped.
Aug 4 11:15:45 chundurur-virtual-machine systemd[1]: Condition check resulted in Process error reports when automatic reporting is enabled (timer based) being skipped.
Aug 4 11:15:45 chundurur-virtual-machine kernel: [ 4.299015] RAS: Correctable Errors collector initialized.
Aug 4 11:15:46 chundurur-virtual-machine alsactl[915]: alsalib main.c:1412:(snd_use_case_mgr_open) error: failed to import hw:0 use case configuration -2
Aug 4 11:15:46 chundurur-virtual-machine networkd-dispatcher[819]: ERROR:Unknown state for Interface NetworkctlListState(idx=1, name='lo', type='loopback', operational='n/a', administrative='unmanaged'):/n/a
Aug 4 11:15:47 chundurur-virtual-machine networkd-dispatcher[819]: ERROR:Unknown state for Interface NetworkctlListState(idx=2, name='ens33', type='ether', operational='n/a', administrative='unmanaged'):/n/a
Aug 4 11:15:48 chundurur-virtual-machine gnome-session-binary[1010]: GLib-GIO-CRITICAL: g_bus_get_sync: assertion 'error == NULL || *error == NULL' failed
Aug 4 11:15:48 chundurur-virtual-machine gnome-session[1010]: gnome-session-binary[1010]: GLib-GIO-CRITICAL: g_bus_get_sync: assertion 'error == NULL || *error == NULL' failed
Aug 4 11:15:48 chundurur-virtual-machine gnome-session[1010]: gnome-session-binary[1010]: GLib-GIO-CRITICAL: g_bus_get_sync: assertion 'error == NULL || *error == NULL' failed
Aug 4 11:15:48 chundurur-virtual-machine gnome-session-binary[1010]: GLib-GIO-CRITICAL: g_bus_get_sync: assertion 'error == NULL || *error == NULL' failed
chundurur@chundurur-virtual-machine:~$ sudo tail -n 50 /var/log/syslog | grep -i "error"
Aug 8 17:34:21 chundurur-virtual-machine gnome-shell[1640]: 35 ERROR: Failed to Initialize Fprintd service: Gio.IOErrorEnum: GDBus.Error:net.reactivated.Fprint.Error.NoSuchDevice: No devices available#01
asynccallback@resource:///org/gnome/gjs/modules/core/overrides/Gio.js:114:23
chundurur@chundurur-virtual-machine:~$

```

Purpose

The goal of setting up a web server like Apache or Nginx is to host web applications and deliver static or dynamic web content to clients over HTTP or HTTPS.

Command Execution and Verification

Installing and starting a web server involves running system package commands like `sudo apt install nginx` or `sudo apt install apache2`, followed by `sudo systemctl start nginx` to initiate the service. Running `sudo systemctl status nginx` or accessing the server IP address in a web browser verifies that the web server is active and serving web pages correctly.

Conclusion

Deploying a web server provides the necessary infrastructure to manage web traffic, process HTTP requests, and serve application content reliably across a network.

26. Set up a web server (e.g., Apache or Nginx).

```

chundurur@chundurur-virtual-machine:~$ sudo systemctl status apache2
apache2.service - The Apache HTTP Server
Loaded: loaded (/lib/systemd/system/apache2.service; enabled; vendor preset: enabled)
Active: failed (Result: exit-code) since Sat 2026-08-08 15:41:35 IST; 2h 37min ago
Docs: https://httpd.apache.org/docs/2.4/
Process: 9260 ExecStart=/usr/sbin/apachectl start (code=exited, status=1/FAILURE)
CPU: 32ms

Aug 08 15:41:35 chundurur-virtual-machine apachectl[9264]: AH00558: apache2: Could not reliably determine the server's fully qualified domain name, using 127.0.1.1. Set the 'ServerName' directive global
Aug 08 15:41:35 chundurur-virtual-machine apachectl[9264]: (98)Address already in use: AH00072: make_sock: could not bind to address [::]:80
Aug 08 15:41:35 chundurur-virtual-machine apachectl[9264]: (98)Address already in use: AH00072: make_sock: could not bind to address 0.0.0.0:80
Aug 08 15:41:35 chundurur-virtual-machine apachectl[9264]: no listening sockets available, shutting down
Aug 08 15:41:35 chundurur-virtual-machine apachectl[9264]: AH00015: Unable to open logs
Aug 08 15:41:35 chundurur-virtual-machine apachectl[9260]: Action 'start' failed.
Aug 08 15:41:35 chundurur-virtual-machine apachectl[9260]: The Apache error log may have more information.
Aug 08 15:41:35 chundurur-virtual-machine systemd[1]: apache2.service: Control process exited, code=exited, status=1/FAILURE
Aug 08 15:41:35 chundurur-virtual-machine systemd[1]: apache2.service: Failed with result 'exit-code'.

chundurur@chundurur-virtual-machine:~$ sudo systemctl status nginx
nginx.service - A high performance web server and a reverse proxy server
Loaded: loaded (/lib/systemd/system/nginx.service; enabled; vendor preset: enabled)
Active: active (running) since Sat 2026-08-08 15:26:22 IST; 2h 58min ago
Docs: man:nginx(8)
Main PID: 8389 (nginx)
Tasks: 3 (limit: 2202)
Memory: 888.0K
CPU: 64ms
CGroup: /system.slice/nginx.service
├─8389 "nginx: master process /usr/sbin/nginx -g daemon on; master_process on;"
├─8390 "nginx: worker process"
└─8391 "nginx: worker process"

Aug 08 15:26:22 chundurur-virtual-machine systemd[1]: Starting A high performance web server and a reverse proxy server...
Aug 08 15:26:22 chundurur-virtual-machine systemd[1]: Started A high performance web server and a reverse proxy server.

```

Purpose

Setting up a web server like Apache or Nginx enables hosting web pages and applications by servicing incoming HTTP/HTTPS requests on specific network ports.

Command Execution and Verification

Running `sudo systemctl status apache2` shows a failed state because port 80 is already in use (Address already in use: AH00072: make_sock: could not bind to address 0.0.0.0:80). In contrast, running `sudo systemctl status nginx` shows active (running) with a master process and worker processes, confirming Nginx is currently handling port 80 successfully.

Conclusion

Only one web server can bind to port 80 at a time, so starting Apache while Nginx is already active causes a port collision error until one service is stopped or reconfigured to a different port.

27 Configure and secure a MySQL Database


```

chundurur@chundurur-virtual-machine:~$ sudo systemctl status mysql
● mysql.service - MySQL Community Server
   Loaded: loaded (/lib/systemd/system/mysql.service; enabled; vendor preset: enabled)
   Active: active (running) since Sat 2026-08-08 18:41:31 IST; 46s ago
     Process: 15050 ExecStartPre=/usr/share/mysql/mysql-systemd-start pre (code=exited, status=0/SUCCESS)
    Main PID: 15058 (mysqld)
      Status: "Server is operational"
        Tasks: 38 (limit: 2202)
       Memory: 364.9M
          CPU: 1.974s
      CGroup: /system.slice/mysql.service
              └─15058 /usr/sbin/mysqld

Aug 08 18:41:30 chundurur-virtual-machine systemd[1]: Starting MySQL Community Server...
Aug 08 18:41:31 chundurur-virtual-machine systemd[1]: Started MySQL Community Server.
chundurur@chundurur-virtual-machine:~$ sudo mysql_secure_installation

```

```

Remove anonymous users? (Press y|Y for Yes, any other key for No) : Y
Success.

Normally, root should only be allowed to connect from
'localhost'. This ensures that someone cannot guess at
the root password from the network.

Disallow root login remotely? (Press y|Y for Yes, any other key for No) : Y
Success.

By default, MySQL comes with a database named 'test' that
anyone can access. This is also intended only for testing,
and should be removed before moving into a production
environment.

Remove test database and access to it? (Press y|Y for Yes, any other key for No) : Y
- Dropping test database...
Success.

- Removing privileges on test database...
Success.

Reloading the privilege tables will ensure that all changes
made so far will take effect immediately.

Reload privilege tables now? (Press y|Y for Yes, any other key for No) : Y
Success.

All done!

```

Purpose

Setting up a web server like Apache or Nginx enables hosting web pages and applications by servicing incoming HTTP/HTTPS requests on specific network ports.

Command Execution and Verification

Running `sudo systemctl status apache2` shows a failed state because port 80 is already in use (Address already in use: AH00072: make_sock: could not bind to address 0.0.0.0:80). In contrast, running `sudo systemctl status nginx` shows active (running) with a master process and worker processes, confirming Nginx is currently handling port 80 successfully.

Conclusion

Only one web server can bind to port 80 at a time, so starting Apache while Nginx is already active causes a port collision error until one service is stopped or reconfigured to a different port.

28 .Set up a Application Server (e.g.,Apache Tomcat)

```
chundurur@chundurur-virtual-machine:~$ sudo systemctl status tomcat9
● tomcat9.service - Apache Tomcat 9 Web Application Server
   Loaded: loaded (/lib/systemd/system/tomcat9.service; enabled; vendor preset: enabled)
   Active: active (running) since Sat 2022-08-08 19:04:30 IST; 28s ago
     Docs: https://tomcat.apache.org/tomcat-9.0-doc/index.html
   Process: 18204 ExecStartPre=/usr/libexec/tomcat9/tomcat-update-policy.sh (code=exited, status=0/SUCCESS)
    Main PID: 18209 (java)
       Tasks: 34 (limit: 2202)
      Memory: 102.9M
         CPU: 7.157s
    CGroup: /system.slice/tomcat9.service
            └─18209 /usr/lib/jvm/default-java/bin/java -Djava.util.logging.config.file=/var/lib/tomcat9/conf/logging.properties -Djava.util.logging.manager=org.apache.juli.ClassLoad

Aug 08 19:04:33 chundurur-virtual-machine tomcat9[18209]: OpenSSL successfully initialized [OpenSSL 3.0.2 15 Mar 2022]
Aug 08 19:04:34 chundurur-virtual-machine tomcat9[18209]: Initializing ProtocolHandler ["http-nio-8080"]
Aug 08 19:04:34 chundurur-virtual-machine tomcat9[18209]: Server initialization in [2602] milliseconds
Aug 08 19:04:35 chundurur-virtual-machine tomcat9[18209]: Starting service [Catalina]
Aug 08 19:04:35 chundurur-virtual-machine tomcat9[18209]: Starting Servlet engine: [Apache Tomcat/9.0.58 (Ubuntu)]
Aug 08 19:04:35 chundurur-virtual-machine tomcat9[18209]: Deploying web application directory [/var/lib/tomcat9/webapps/ROOT]
Aug 08 19:04:37 chundurur-virtual-machine tomcat9[18209]: At least one JAR was scanned for TLDs yet contained no TLDs. Enable debug logging for this logger for a complete list of JARs
Aug 08 19:04:37 chundurur-virtual-machine tomcat9[18209]: Deployment of web application directory [/var/lib/tomcat9/webapps/ROOT] has finished in [1,980] ms
Aug 08 19:04:37 chundurur-virtual-machine tomcat9[18209]: Starting ProtocolHandler ["http-nio-8080"]
Aug 08 19:04:37 chundurur-virtual-machine tomcat9[18209]: Server startup in [2251] milliseconds
lines 1-22/22 (END)
```

Purpose

Setting up an application server like Apache Tomcat allows Java-based web applications and servlets to run and serve dynamic content over HTTP.

Command Execution and Verification

The command `sudo systemctl status tomcat9` checks the status of the Tomcat service, returning active (running) with process logs showing initializing ProtocolHandler ["http-nio-8080"] and Server startup in [2251] milliseconds. This verifies that Apache Tomcat is successfully initialized and running on port 8080.

Conclusion

Deploying an application server requires verifying the service state via `systemctl` to confirm that the Java runtime environment and web application directories are deployed and listening on the designated port.

29 ,create a service file for Apache Tomcat.(Should execute by using `systemctl` command)


```

chundurur@chundurur-virtual-machine:~$ sudo systemctl status tomcat9
● tomcat9.service - Apache Tomcat 9 Web Application Server
   Loaded: loaded (/lib/systemd/system/tomcat9.service; enabled; vendor preset: enabled)
   Active: active (running) since Sat 2026-08-08 19:04:30 IST; 20s ago
     Docs: https://tomcat.apache.org/tomcat-9.0-doc/index.html
   Process: 18204 ExecStartPre=/usr/libexec/tomcat9/tomcat-update-policy.sh (code=exited, status=0/SUCCESS)
   Main PID: 18209 (java)
    Tasks: 34 (limit: 2202)
   Memory: 102.9M
      CPU: 7.157s
   CGroup: /system.slice/tomcat9.service
           └─18209 /usr/lib/jvm/default-java/bin/java -Djava.util.logging.config.file=/var/lib/tomcat9/conf/logging.properties -Djava.util.logging.manager=org.apache.juli.ClassLoad
Aug 08 19:04:33 chundurur-virtual-machine tomcat9[18209]: OpenSSL successfully initialized [OpenSSL 3.0.2 15 Mar 2022]
Aug 08 19:04:34 chundurur-virtual-machine tomcat9[18209]: Initializing ProtocolHandler ["http-nio-8080"]
Aug 08 19:04:34 chundurur-virtual-machine tomcat9[18209]: Server initialization in [2602] milliseconds
Aug 08 19:04:35 chundurur-virtual-machine tomcat9[18209]: Starting service [Catalina]
Aug 08 19:04:35 chundurur-virtual-machine tomcat9[18209]: Starting Servlet engine: [Apache Tomcat/9.0.58 (Ubuntu)]
Aug 08 19:04:35 chundurur-virtual-machine tomcat9[18209]: Deploying web application directory [/var/lib/tomcat9/webapps/ROOT]
Aug 08 19:04:37 chundurur-virtual-machine tomcat9[18209]: At least one JAR was scanned for TLDs yet contained no TLDs. Enable debug logging for this logger for a complete list of JARs
Aug 08 19:04:37 chundurur-virtual-machine tomcat9[18209]: Deployment of web application directory [/var/lib/tomcat9/webapps/ROOT] has finished in [1,980] ms
Aug 08 19:04:37 chundurur-virtual-machine tomcat9[18209]: Starting ProtocolHandler ["http-nio-8080"]
Aug 08 19:04:37 chundurur-virtual-machine tomcat9[18209]: Server startup in [2251] milliseconds
lines 1-22/22 (END)

```

Purpose

Setting up an application server like Apache Tomcat allows Java-based web applications and servlets to run and serve dynamic content over HTTP.

Command Execution and Verification

The command `sudo systemctl status tomcat9` checks the status of the Tomcat service, returning active (running) with process logs showing initializing ProtocolHandler ["http-nio-8080"] and Server startup in [2251] milliseconds. This verifies that Apache Tomcat is successfully initialized and running on port 8080.

Conclusion

Deploying an application server requires verifying the service state via `systemctl` to confirm that the Java runtime environment and web application directories are deployed and listening on the designated port.

30 .Print specific columns from a delimited file.

```

chundurur@chundurur-virtual-machine:~$ mkdir f1
chundurur@chundurur-virtual-machine:~$ cd f1
chundurur@chundurur-virtual-machine:~/f1$ touch f2
chundurur@chundurur-virtual-machine:~/f1$ vi f2
chundurur@chundurur-virtual-machine:~/f1$ cut -d',' -f1,3 f2
Cbash,58
bash,55
kk,88
chundurur@chundurur-virtual-machine:~/f1$

```

Purpose

Printing specific fields or columns from a delimited file allows users to extract only relevant data fields from structured text files like CSVs.

Command Execution and Verification

The command `cut -d',' -f1,3 f2` reads file `f2` using comma as the delimiter and extracts fields 1 and 3, outputting rows like `Cbash,58`, `bash,55`, and `kk,88`. This verifies that only the specified columns are selected and printed to stdout.

Conclusion

Using the `cut` command with `-d` to define the delimiter and `-f` to specify field numbers provides an efficient way to parse and extract targeted columns from structured tabular data.

31 .Filter and print lines based on a specific pattern or condition.

```
chundurur@chundurur-virtual-machine:~$ grep "apple" test.text
apple
apple juice
chundurur@chundurur-virtual-machine:~$ grep -i "apple" test.text
apple
apple juice
chundurur@chundurur-virtual-machine:~$ grep -v "apple" test.text
banana
orange
grapes
chundurur@chundurur-virtual-machine:~$
```

Purpose

Filtering text using `grep` allows users to search for specific character patterns or keywords within a file and display matching or non-matching lines.

Command Execution and Verification

The command `grep "apple" test.text` searches for the exact string `apple`, while `grep -i "apple" test.text` performs a case-insensitive search to match terms like `Apple` or `APPLE`, both outputting matching lines such as `apple` and `apple juice`. Running `grep -v "apple" test.text` inverts the search to exclude lines containing `apple`, printing only `banana`, `orange`, and `grapes`.

Conclusion

The `grep` command provides versatile pattern matching, using flags like `-i` to ignore case sensitivity and `-v` to filter out unwanted lines from textual output.

32 Calculate and print the average, sum, or other statistics of a column.

```

chunduru@chunduru-virtual-machine:~$ nano marks.txt
chunduru@chunduru-virtual-machine:~$ awk '{sum += $2} END {print "sum =",sum}' marks.txt
awk: 1: unexpected character '&'
chunduru@chunduru-virtual-machine:~$ wk '{sum += $2} END {print "sum =",sum}' marks.txt
wk: command not found
chunduru@chunduru-virtual-machine:~$ ^[[200~wk '{sum += $2} END {print "sum =",sum}' marks.txt
wk: command not found
chunduru@chunduru-virtual-machine:~$ ~^C
chunduru@chunduru-virtual-machine:~$ awk '{sum += $2} END {print "sum =",sum}' marks.txt
sum = 300
chunduru@chunduru-virtual-machine:~$ awk '{sum += $2; count++} END {print "Average =", sum/count}' marks.txt
Average = 60
chunduru@chunduru-virtual-machine:~$

```

Purpose

The primary goal of using awk in this context is to perform statistical calculations, such as summing values and calculating the average across a specific column of data in a text file.

Command Execution and Verification

The command `awk '{sum += $2} END {print "sum =", sum}' marks.txt` iterates through each line of `marks.txt`, adds the values in column 2, and outputs `sum = 300` upon completion. To find the mean, `awk '{sum += $2; count++} END {print "Average =", sum/count}' marks.txt` tracks both total sum and line count, successfully outputting `Average = 60`.

Conclusion

Using awk with column variable referencing and an END block provides an efficient way to perform mathematical calculations and generate summary metrics directly from structured text data.

33 Perform string manipulation, such as extracting substrings or changing case.

```

chunduru@chunduru-virtual-machine:~$ nano marks.txt
chunduru@chunduru-virtual-machine:~$ awk '{sum += $2} END {print "sum =",sum}' marks.txt
awk: 1: unexpected character '&'
chunduru@chunduru-virtual-machine:~$ wk '{sum += $2} END {print "sum =",sum}' marks.txt
wk: command not found
chunduru@chunduru-virtual-machine:~$ ^[[200~wk '{sum += $2} END {print "sum =",sum}' marks.txt
wk: command not found
chunduru@chunduru-virtual-machine:~$ ~^C
chunduru@chunduru-virtual-machine:~$ awk '{sum += $2} END {print "sum =",sum}' marks.txt
sum = 300
chunduru@chunduru-virtual-machine:~$ awk '{sum += $2; count++} END {print "Average =", sum/count}' marks.txt
Average = 60
chunduru@chunduru-virtual-machine:~$ name="chunduru"
chunduru@chunduru-virtual-machine:~$ echo "$name"
chunduru
chunduru@chunduru-virtual-machine:~$ echo "${name:0:4}"
> ^C
chunduru@chunduru-virtual-machine:~$ echo "${name :0:4}"
-bash: ${name :0:4}: bad substitution
chunduru@chunduru-virtual-machine:~$ echo "${name:0:4}"
chun
chunduru@chunduru-virtual-machine:~$ echo "${#name}"
8
chunduru@chunduru-virtual-machine:~$ echo "$name" | tr '[:upper;]' '[:lower;]'
chndrlrl
chunduru@chunduru-virtual-machine:~$ echo "Hello Chunduru " | sed 's/chunduru/world/'
Hello Chunduru
chunduru@chunduru-virtual-machine:~$

```

Purpose

Performing string manipulation in Bash allows users to extract substrings, check variable lengths, and transform text cases using parameter expansion and stream editors like tr and sed.

Command Execution and Verification

Setting name="chunduru" and executing echo "\${name:0:4}" extracts the first four characters, outputting chun, while echo "\${#name}" measures and prints the string length of 8. Case conversion using echo "\$name" | tr '[:upper:]' '[:lower:]' converts all characters to lowercase (chunduru), and running echo "Hello Chunduru" | sed 's/chunduru/world/' attempts pattern replacement but outputs Hello Chunduru unchanged due to case sensitivity.

Conclusion

Bash parameter expansion provides efficient built-in substring extraction and length calculation, while external tools like tr and sed extend text manipulation capabilities for case conversion and string substitution.

34 Count the occurrences of a specific pattern in a file.

```
chunduru@chunduru-virtual-machine:~$ grep -c "apple" test.text
3
chunduru@chunduru-virtual-machine:~$ grep -oi "apple" test.text | wc -l
3
chunduru@chunduru-virtual-machine:~$ g
```

Purpose

Counting occurrences of a specific pattern in a file allows users to determine how many lines or total individual instances contain a target keyword or phrase.

Command Execution and Verification

The command grep -c "apple" test.text counts matching lines and outputs 3, verifying that there are three lines containing the string "apple". Running grep -oi "apple" test.text | wc -l uses -o to extract every individual match on a separate line and -i to ignore case, piping the results to wc -l to verify a total count of 3 occurrences across the file.

Conclusion

Using grep -c provides a quick count of lines matching a pattern, while piping grep -oi into wc -l counts every individual instance regardless of line structure or letter casing.

35 Sort lines based on a specific field or column.

```

chunduru@chunduru-virtual-machine:~$ sort -k2 -n marks.txt
anil 60
kiran 70
ravi 80
suresh 90
chunduru@chunduru-virtual-machine:~$ sort -k2 -nr marks.txt
suresh 90
ravi 80
kiran 70
anil 60
chunduru@chunduru-virtual-machine:~$ █

```

Purpose

Sorting data in structured files allows users to order records numerically or alphabetically based on a specific key or column value.

Command Execution and Verification

The command `sort -k2 -n marks.txt` sorts the lines in `marks.txt` numerically using the second column (`-k2 -n`), displaying records in ascending order from 60 to 90 (anil 60 to suresh 90). Running `sort -k2 -nr marks.txt` adds the `-r` flag to reverse the sort order, successfully outputting the records in descending order from 90 to 60 (suresh 90 to anil 60).

Conclusion

Using `sort` with the `-k` option targets specific columns for organization, while combining flags like `-n` for numerical sorting and `-r` for reverse ordering enables precise control over data sequence.

36 Merge multiple files based on a common field or column.

```

chunduru@chunduru-virtual-machine:~$ nano file1.txt
chunduru@chunduru-virtual-machine:~$ nano file2.txt
chunduru@chunduru-virtual-machine:~$ join file1.txt file2.txt
join: file1.txt: No such file or directory
chunduru@chunduru-virtual-machine:~$ ls
1      Documents  extract  file1.txt  file.txt  lin1.txt  Pictures  snap  Templates  test-tar  test.tar.gz  Videos
Desktop Downloads  f3      file2.txt  karthik  Music     Public   techie.tar  test     test.tar  test.text
chunduru@chunduru-virtual-machine:~$ join file1.txt file2.txt
join: file2.txt:4: is not sorted:
join: file1.txt:2: is not sorted: kiran
join: input is not in sorted order
chunduru@chunduru-virtual-machine:~$ sort -k1 -n file1.txt -o file1.txt
chunduru@chunduru-virtual-machine:~$ sort -k1 -n file2.txt -o file2.txt
chunduru@chunduru-virtual-machine:~$ join file1.txt file2.txt

```

Purpose

Merging files using the `join` command combines lines from two text files that share a common key or field value.

Command Execution and Verification

The command `join file1.txt file2.txt` initially fails because `join` requires input files to be sorted numerically or alphabetically beforehand. To fix this, running `sort -k1 -n file1.txt -o file1.txt` and `sort -k1 -n file2.txt -o file2.txt` sorts both files in place by their first field. Re-running `join file1.txt file2.txt` successfully matches corresponding records based on column 1 and outputs the combined dataset.

Conclusion

The join command provides an efficient way to merge datasets on a shared key, provided both input files are pre-sorted along the join field using sort.

37 Substitute text in a file using search and replace.

```
chunduru@chunduru-virtual-machine:~$ nano test.txt
chunduru@chunduru-virtual-machine:~$ sed 's/ravi/kiran/g' test.txt
hello kiran
hello kiran
welcome kiran

chunduru@chunduru-virtual-machine:~$ sed -i 's/ravi/kiran/g' test.txt
chunduru@chunduru-virtual-machine:~$ cat test.txt
hello kiran
hello kiran
welcome kiran

chunduru@chunduru-virtual-machine:~$
```

Purpose

Replacing or substituting text within a file allows users to swap specific words or phrases dynamically without manually editing the file line by line. Command Execution and Verification

The command `sed 's/ravi/kiran/g' test.txt` searches for all instances of "ravi" in test.txt and replaces them with "kiran", displaying the modified text in stdout while leaving the original file unchanged. Running `sed -i 's/ravi/kiran/g' test.txt` applies the -i flag to perform the substitution in-place directly on test.txt, which is verified by running `cat test.txt` to display updated lines containing "hello kiran" and "welcome kiran".

Conclusion

Using sed with the `s/pattern/replacement/g` syntax performs global search-and-replace transformations, while adding the -i option permanently saves those modifications directly to the source file.

38 Delete specific lines based on a pattern or line number.

```
chunduru@chunduru-virtual-machine:~$ sed 's/ravi/kiran/g' test.txt
hello kiran
hello kiran
welcome kiran

chunduru@chunduru-virtual-machine:~$ sed -i 's/ravi/kiran/g' test.txt
chunduru@chunduru-virtual-machine:~$ cat test.txt
hello kiran
hello kiran
welcome kiran

chunduru@chunduru-virtual-machine:~$ sed -i '/ravi/d' test.txt
chunduru@chunduru-virtual-machine:~$ cat test.txt
hello kiran
hello kiran
welcome kiran

chunduru@chunduru-virtual-machine:~$ sed -i '2d' test.txt
chunduru@chunduru-virtual-machine:~$ cat test.txt
hello kiran
welcome kiran

chunduru@chunduru-virtual-machine:~$ sed -i '2,4d' test.txt
chunduru@chunduru-virtual-machine:~$ cat test.txt
hello kiran

chunduru@chunduru-virtual-machine:~$
```


Purpose

Deleting specific lines in a file using sed allows users to remove unwanted content based on matching text patterns or exact line numbers.

Command Execution and Verification

The command `sed -i '/ravi/d' test.txt` deletes all lines containing the pattern "ravi", while `sed -i '2d' test.txt` removes line 2 specifically from the file. Running `sed -i '2,4d' test.txt` deletes a range of lines from line 2 through line 4 inclusive, which is verified using `cat test.txt` to display the remaining lines in the file.

Conclusion

Using sed with the d address command provides precise control for line removal, supporting pattern matching, single line deletion, and multi-line range removal directly within source files.

39Append or insert text before or after a specific pattern or line.

```
chundurur@chundurur-virtual-machine:~$ sed -i '/ravi/welcome' test.txt
chundurur@chundurur-virtual-machine:~$ cat test.txt

hello kiran
ravi
suresh

chundurur@chundurur-virtual-machine:~$ sed -i '/ravi/i Welcome' test.txt
chundurur@chundurur-virtual-machine:~$ cat test.txt

hello kiran
Welcome
ravi
suresh

chundurur@chundurur-virtual-machine:~$ sed -i '2a Hello' test.txt
chundurur@chundurur-virtual-machine:~$ cat test.txt

hello kiran
Hello
Welcome
ravi
suresh

chundurur@chundurur-virtual-machine:~$
```

Purpose

Appending or inserting text in a file using sed allows users to place new lines above or below specific matching patterns or specified line numbers.

Command Execution and Verification

The command `sed -i '/ravi/i Welcome' test.txt` inserts the text "Welcome" on a new line immediately before the line matching "ravi", while `sed -i '2a Hello' test.txt` appends "Hello" on a new line directly after line 2. Running `cat test.txt` verifies that the new lines "Welcome" and "Hello" are added at the precise target locations in the file.

Conclusion

Using sed with the `i` command inserts content before a target line, whereas using the `a` command appends content after a target line, offering flexible line-level text insertion.

40 Print only specific lines from a file.

```
chundurur@chundurur-virtual-machine:~$ sed -n '2p' test.txt
hello kiran
chundurur@chundurur-virtual-machine:~$ sed -n '2,4d' test.txt
chundurur@chundurur-virtual-machine:~$ cat test.txt

hello kiran
Hello
Welcome
ravi
suresh

chundurur@chundurur-virtual-machine:~$ head -n 1 test.txt
hello kiran
chundurur@chundurur-virtual-machine:~$ tail -n 1 test.txt
suresh
chundurur@chundurur-virtual-machine:~$
```

Purpose: To learn and demonstrate basic Linux text-processing commands like sed, head, tail, and cat.

Commands: `sed -n '2p' test.txt`, `sed -n '2,4d' test.txt`, `cat test.txt`, `head -n 1 test.txt`, `tail -n 1 test.txt`.

Conclusion: These commands help display, delete, and view specific lines or portions of a text file efficiently.

41 Copy file from linux to windows machine

42 .5 use cases for AWK and 5 use cases for sed

```

chundurur@chundurur-virtual-machine:~$ awk '{print $s}' test.txt
hello kiran
Hello
Welcome
ravi
suresh

chundurur@chundurur-virtual-machine:~$ awk '{sum += $2} END {print sum}' test.txt
9

chundurur@chundurur-virtual-machine:~$ awk '{sum += $2; count++} END {print sum/count}' test.txt
awk: cannot open text.txt (No such file or directory)
chundurur@chundurur-virtual-machine:~$ awk '$2 > 50 {print $1 , $2}' test.txt
awk: cannot open text.txt (No such file or directory)
chundurur@chundurur-virtual-machine:~$ awk '$2 > 50 {print $1, $2}' test.txt
hello kiran
chundurur@chundurur-virtual-machine:~$ awk 'END {print NR}' test.txt
7

chundurur@chundurur-virtual-machine:~$ sed -i 's/old/new/g' test.txt
chundurur@chundurur-virtual-machine:~$ cat test.txt

hello kiran
Hello
Welcome
ravi
suresh

chundurur@chundurur-virtual-machine:~$ sed -i '2d' test.txt
chundurur@chundurur-virtual-machine:~$ cat test.txt

Hello
Welcome
ravi
suresh

```

```

chundurur@chundurur-virtual-machine:~$ sed -i '2d' test.txt
chundurur@chundurur-virtual-machine:~$ cat test.txt

Hello
Welcome
ravi
suresh

chundurur@chundurur-virtual-machine:~$ sed -i '/ravi/d' test.txt
chundurur@chundurur-virtual-machine:~$ cat test.txt

Hello
Welcome
suresh

chundurur@chundurur-virtual-machine:~$ sed -i '2,4p' test.txt
chundurur@chundurur-virtual-machine:~$ cat test.txt

Hello
Hello
Welcome
Welcome
suresh
suresh

chundurur@chundurur-virtual-machine:~$ sed -i '2i Hello' test.txt
chundurur@chundurur-virtual-machine:~$ cat test.txt

Hello
Hello
Hello
Welcome
Welcome
suresh
suresh

chundurur@chundurur-virtual-machine:~$

```

Purpose: To practice awk and sed commands for displaying, modifying, deleting, counting, and processing text in a file.

Commands: awk, sed -i 's/old/new/g', sed -i '2d', sed -i '/ravi/d', sed -i '2,4p', and sed -i '2i Hello'.

Conclusion: awk and sed are powerful Linux utilities used to efficiently search, edit, filter, and manipulate text files.